



Fig. 8: Prototype set up in EFY Lab

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at source.efymag.com

TABLE II
Test Points

Test point	Details
TP0	0V (GND)
TP1	5V
TP2	Low when strip is in the slotted gap of IC1
TP3	Low when S2 is pressed

The prototype set up in EFY Lab is shown in Fig. 8.

For troubleshooting, first check the voltages at various test points as listed in Table II.

Software program

The program for the microcontroller is written in C language and compiled using AVR Studio 4. The compiler is used to convert C program into a hex file for burning into the flash memory of ATmega128 microcontroller using AVR Studio 4. A 4-pin connector (CON2) is provided in the circuit for programming the chip using a suitable ISP-compatible AVR programmer board.

The complete program is a combination of different functions. It starts with the main function, which

is used to configure various ports of ATmega128 microcontroller as input/output (I/O) and initialise the LCD. It also calls UAR_Tinit() function to initialise UART0 to 9600bps and enable reception. The main function also enables interrupts to be used in the project and continuously rotates

the motor in either direction (clockwise or anti-clockwise) at 2rpm.

The function `angle_disp()` is used for displaying the corresponding angle of the rotation of ultrasonic sensor module from its point of origin. There are two interrupt handler sub-routine functions, `ISR(UART0_RX_vect)` and `ISR(INT4_vect)`, which represent operation mode on receiving interrupts.

`ISR(INT4_vect)` is used to set reference for the angle and operation of the motor in the opposite direction to the current direction of the rotation of the motor. `ISR(UART0_RX_vect)` receives nine bytes from the sensor, calculates the distance of the object from the sensor, and displays it on the LCD screen. As soon as an object is detected, the buzzer sounds and indicates the angle of object detection on the LCD as well as on the LED dial with corresponding LED indication.

All LCD functions, such as `lcd_init()`, `lcd_cmd()`, `lcd_str()` and `lcd_num()`, are included in LCD header file `lcd.h`. ●

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